

and Galvani (2018) also find a similar result in the corporate bond market.

One effective way to examine time series reversal in financial assets is to employ contrarian trend-following strategies. Trend-following trading signals depend on their own past returns without a cross-sectional comparison. The simple moving average is one of the most intuitive trend-following strategies based on technical analysis. Recently, researchers have attributed the profitability of trend-following strategies to time series continuation and introduced more advanced methods such as TSM, see Moskowitz et al. (2012), and return signal momentum (RSM), see Papailias et al. (2021). Therefore, we base our investigation on these two momentum approaches in the time series domain.

We first capture the timing of time series reversal and find that it occurs from 12 to 24 months after the portfolio formation. This finding differs from the traditional cross-sectional reversals which usually last longer, between 2 and 5 years after the portfolio formation date.⁴ Moskowitz et al. (2012) document a strong short-term (1–12 month) TSM, but no statistically significant long-term (2–5 years) reversal.

Like Conrad and Yavuz (2017), we classify instruments in trend-following strategies into four groups: “realised winner”, “realised loser”, “contrarian winner”, and “contrarian loser”. The realised winner sub-portfolio consists of instruments with positive trend-following trading signals (winners) whose profits are realised in the subsequent investment horizon (1–12 months). The realised loser sub-portfolio consists of instruments with negative trading signals (losers) that later produce negative returns, meaning that the losers portfolio realises its profit. By contrast, the contrarian winners and contrarian losers include instruments with positive and negative signals that fail to gain profits. The construction of the above four sub-portfolios allows us to see which part of the trend-following returns contributes most to time

⁴Of course, we cannot rule out the possibility that this difference might occur due to various factors such as the sample period and the specific market. For example, cross-sectional reversals require a long-term ranking period of 2–5 years in stock markets (De Bondt and Thaler, 1985), but reduce to 1.5–3 years in commodity markets, as suggested by Bianchi et al. (2015). Bhojraj and Swaminathan (2006) find that the reversals of country indices occur from 2 to 3 years after the 1-year momentum effect.